higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOFWORK
NQF LEVEL 3

21 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1: HIPPED ROOFS

1.1
1.1.1 False
1.1.2 True
1.1.3 True
1.1.4 False
1.1.5 False
1.1.6 False
1.1.7 True
1.1.8 False
(8 × 1) (8)

1.2
- Structural stability
- Safety margins
- Economy (3)

1.3
- Timber was cut too short
- Timber was not cut square
- Splitting timber due to poor nailing practice
- Inability to measure accurately (Any 2 × 1) (2)

1.4 To ensure that all the chords and webs✓ align perfectly✓ (2)

1.5
- 12 mm bolts
- 100 mm nails (2)

1.6
1.6.1 HG – hip girder
1.6.2 GT – girder truss
1.6.3 BC – bottom chord runner
(3 × 1) (3)

1.7 If there are no soffits, the bricklayer✓ must be given a chance to beam-fill✓ between the rafters✓ because this cannot be done afterwards✓ (4)

1.8
- The roof must be designed by a professional
- Additional metal brackets and truss hangers are required
- Skilled workers must be used
- Roof covering material which must be cut results in wastage (4)

1.9
- At the wall plate
- At the hip (2)

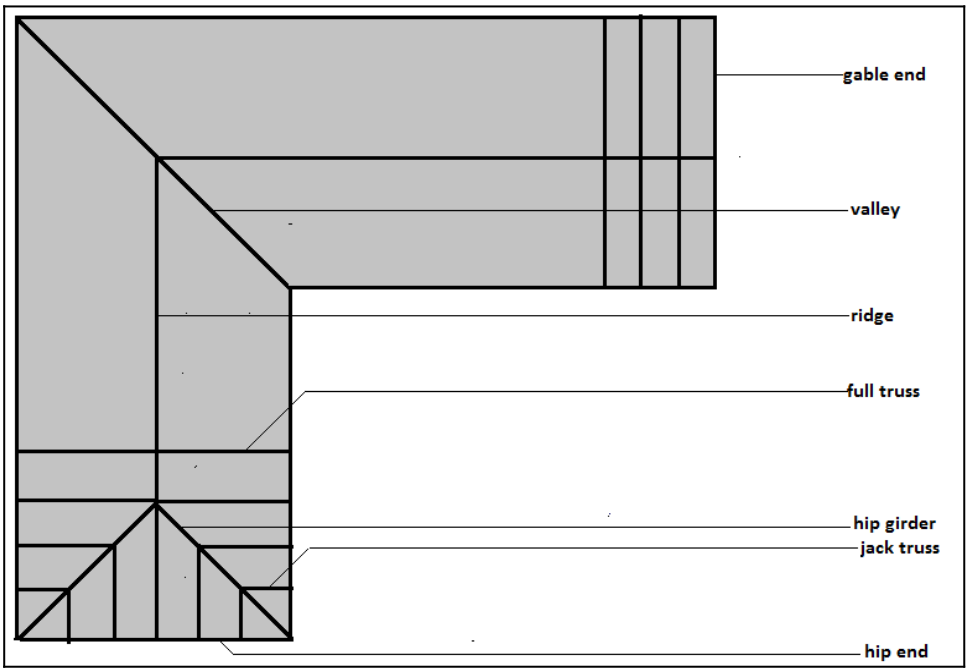
[30]

QUESTION 2: ERECT AN L-SHAPED ROOF

2.1 2.1.1 double wedges
2.1.2 wall plate
2.1.3 tiling batten
2.1.4 safety harness
2.1.5 diagonal bracing

(5 × 1) (5)

2.2



2.2.1 The gable end (correctly drawn and labelled)
2.2.2 The hip end (correctly drawn and labelled)
2.2.3 A hip girder (correctly drawn and labelled)
2.2.4 A jack truss (correctly drawn and labelled)
2.2.5 The valley (correctly drawn and labelled)
2.2.6 The ridge (correctly drawn and labelled)
2.2.7 A full truss (correctly drawn and labelled)

(2 × 7) (14)

2.3
- A valley carries a bigger water load than the rest of the roof
- Leaves and debris can get stuck in the valley and create a blockage
- Poor workmanship could result in a weak valley (3)

2.7 **Step 1**: Keep the truss furthest from the valley about 50 mm from the gable wall
Step 2: Measure the distance from that truss to the corner of the valley
Step 3: Divide this space by the number of trusses minus one, to obtain the required truss-spacing
Step 4: Do NOT exceed the maximum truss spacing allowed

(2 × 4) (8)
[30]

QUESTION 3: INSPECTION OF ROOF TRUSSES

3.1 3.1.1 The owner must appoint a qualified person to inspect the roof

3.1.2 The roof erector has to ensure that the roof is constructed in accordance with the relevant building regulations

3.1.3 The occupant of the house must report any latent defects to the builder

3.1.4 The designing engineer designs and takes responsibility for the roof

(2 × 4) (8)

3.2
- Battens/purlins should be the correct size.
- The masonry should be correctly trimmed to prevent obstructions.
- The truss spacing must comply with the choice of roof covering.
- All nail plates should be fully driven into the timber.
- The roof timber should be treated according to standard regulations.

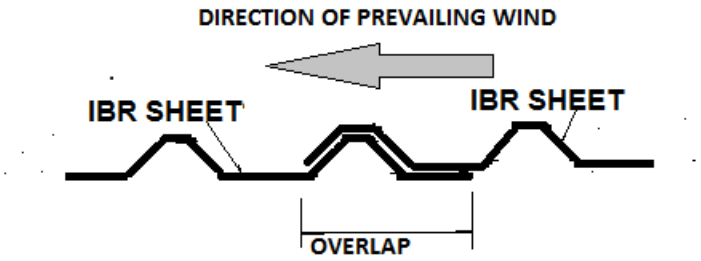
(Any 4 × 1) (4)

3.3 3.3.1 30 millimetres
3.3.2 50 millimetres
3.3.3 50 millimetres

(3 × 1) (3)
[15]

QUESTION 4: CLAD A TIMBER ROOF STRUCTURE

4.1



IBR Sheets	2
Direction of wind	2
Overlap	2
Total	**6**

(6)

4.2	4.2.1	Metal and fibre-cement sheets	(Any ONE)	(1)
	4.2.2	• Concrete tiles • Clay tiles • Natural slate • Thatch	(Any ONE)	(1)
	4.2.3	Metal roof tiles		(1)
4.3		• The centre-to-centre spacing of trusses • The roof slope • The number of bolts and connectors • The batten/purlin size and spacing • The roof span	(Any 4 × 1)	(4)
4.4		• To reduce attic pressure (suction on the tiles caused by strong winds) • To prevent dust accumulation		(2)
4.5		• Type of building • Roof pitch • Wind speed • Length of the roof slope • Terrain category • Height of the roof from the ground to the eaves	(Any 5 × 1)	(5)
4.6		• The reflective properties of bituminous aluminium reduce the temperature inside a building. • It also protects malthoid and roofing felt from ultra-violet degradation (sun damage).	(2 × 2)	(4)
4.7		At the ridge		(1)
				[25]
			TOTAL:	**100**

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